CATS$^+$

Competishun Advance

Training Sessions

(Lecture - VIII)

One

Statement-1 : There lies exactly 3 points on the curve $8x^3 + y^3 + 6xy = 1$, which form an equilateral triangle.

because

Statement-2 : The locus of all point $P(x, y)$ satisfying $8x^3 + y^3 + 6xy = 1$ consists of union of a straight line and a point not on the line.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Sol

$$8x^3 + y^3 + 6xy = 1$$

$$(2x)^3 + (y)^3 + (-1)^3 - 3(2x)(y)(-1) = 0$$

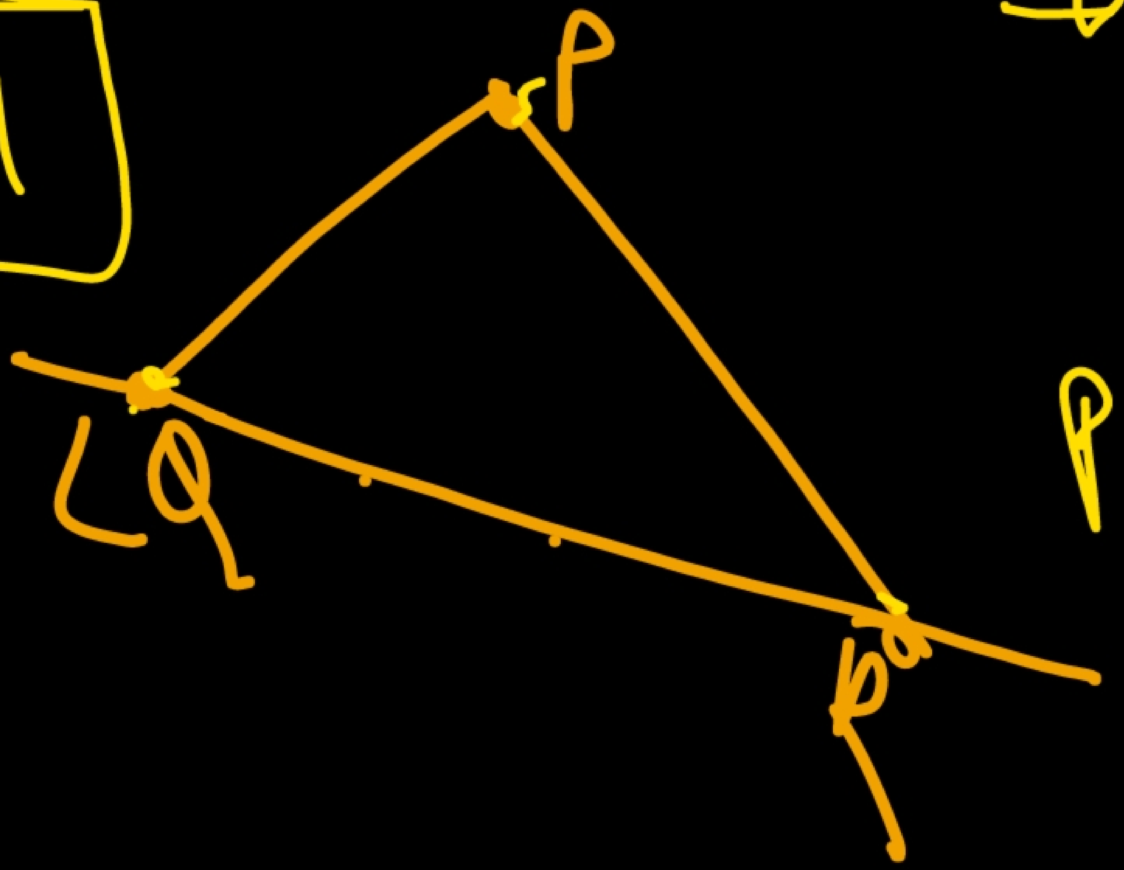
$$2x + y - 1 = 0 \quad \text{or} \quad 2x = y = -1$$

$$\boxed{2x + y = 1}$$

$x = -\frac{1}{2}$

$y = -1$

$P : \left(-\frac{1}{2}, -1\right)$



$$a^3 + b^3 + c^3 - 3abc = 0$$

$$= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$a+b+c = 0$ or $a = b = c$

Que

Let a,b,c are positive real number such that $p = a^2b + ab^2 - a^2c - ac^2$, $q = b^2c + bc^2 - a^2b - ab^2$ and $r = ac^2 + a^2c - cb^2 - bc^2$ and the quadratic equation $px^2 + qx + r = 0$ has equal roots. If $ab + bc - \lambda ac = 0$ then determine λ

Sol

$a, b, c \in R^+$

$p = a^2b + ab^2 - a^2c - ac^2 = a(ab + b^2 - ac - c^2) = a(a(b-c) + (b-c)(b+c))$

$= a(b-c).(a+b+c)$

$q = b, c - - - = b.(c-a).(a+b+c)$

$r = a, c - - - = c.(a-b)(a+b+c)$

$px^2 + qx + r = 0$

$D = 0$

$q^2 = 4pr$

$b^2(c-a)^2\cancel{(a+b+c)^2} = 4.a.c.(a-b)(b-c)\cancel{(a+b+c)^2}$

$b^2(c-a)^2 = 4ac(a-b)(b-c) = 4ac(-b^2 + ab - ac + bc)$

$b^2((c-a)^2 + 4ac) = 4ac(ab + bc - ac)$

$b^2((a+c)^2) = 4ac\,b(a+c) - 4(a^2c^2)$

$ab + bc - \lambda ac = 0$

$b(a+c) = \lambda ac$

Alt: $b(a+c) = \lambda ac$

$$(b(a+c))^2 = 4ac\,b(a+c) - 4a^2c^2$$

$$\cancel{\lambda^2 a^2 c^2} = 4\cancel{ac}\,\lambda \cancel{ac} - 4\cancel{a^2c^2}$$

$$\lambda^2 = 4\lambda - 4$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)^2 = 0$$

$$\lambda = 2$$

$$(b(a+c))^2 + (2ac)^2 - 2\cdot 2ac\cdot b(a+c) = 0$$

$$(b(a+c) - 2ac)^2 = 0$$

$$\Rightarrow b(a+c) - 2ac = 0.$$

$$ab + bc - \underset{\downarrow}{2}ac = 0$$

$$\lambda$$

$$\Rightarrow \boxed{\lambda = 2}$$ Ans

Que

Find the number of quadratic equations whose roots are α,β satisfying $\alpha^2+\beta^2=5$, $3(\alpha^5+\beta^5)=11(\alpha^3+\beta^3)$, where $\alpha,\beta\in R$.

Soln

$\alpha^2+\beta^2=5$ ✓

$3(\alpha^5+\beta^5)=11(\alpha^3+\beta^3)$

$\Rightarrow 3\left((\alpha+\beta)(\alpha^4-\alpha^3\beta+\alpha^2\beta^2-\alpha\beta^3+\beta^4)\right)$

$=11\cdot(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$

C-1 $\alpha+\beta=0$

$\alpha^2+\beta^2=5$

$\alpha^2=5/2$

$\alpha=\pm\sqrt{\frac{5}{2}}$

$\boxed{x^2=5/2}\rightarrow \phi_1$

C-2 $\alpha+\beta\neq 0$

$\Rightarrow 3\left(\alpha^4+\beta^4-\alpha\beta(\alpha^2+\beta^2)+(\alpha\beta)^2\right)=11\cdot(\alpha^2+\beta^2-\alpha\beta)$

$(\alpha^2+\beta^2)^2-2(\alpha\beta)^2$

$\Downarrow$ $\alpha^2+\beta^2=5$

$\alpha\beta=t$

$3\left(25-2t^2-5t+t^2\right)=11(5-t)$

$75-3t^2-15t=55-11t$

$3t^2+4t-20=0$

$t=\frac{-4\pm\sqrt{16+240}}{2\cdot 3}=\frac{-4\pm16}{2\cdot3}=-\frac{10}{3},2$

$$\alpha^2+\beta^2=5$$

$$3(\alpha^5+\beta^5)=11\cdot(\alpha^3+\beta^3)$$

$\alpha=1$
$\beta=2$ ✓

$$3\cdot 33=11\cdot 9$$

3- quadratic equation

$$2x^2-5=0$$

$$x^2-3x+2=0$$

$$x^2+3x+2=0$$

$$\alpha\beta=-\frac{10}{3}$$

$$\alpha^2+\beta^2=5$$

$$(\alpha+\beta)^2=2\alpha\beta+5$$

$$=-\frac{20}{3}+5$$

$$=-5/3$$

×

$$\alpha\beta=2$$

$$(\alpha+\beta)^2=2\alpha\beta+5$$

$$=9$$

$$\alpha+\beta=-3,\ 3$$

$$x^2-3x+2=0$$

$\hookrightarrow 1, 2$

$$x^2+3x+2=0$$

$\hookrightarrow -1, -2$

Que

Find the number of ordered pair (x, y) satisfying the equation $3x^2 + 3y^2 - 4xy + 10x - 10y + 10 = 0$, where $x,y \in R$

Sol

$$3x^2+3y^2-4xy+10x-10y+10=0.$$

$$\Rightarrow 3x^2 + 2(5-2y)x + (3y^2-10y+10) = 0$$

$$x = \frac{-\cancel{2}(5-2y) \pm \sqrt{\cancel{4}(5-2y)^2 - \cancel{4}\cdot 3\cdot(3y^2-10y+10)}}{\cancel{2}\cdot 3}$$

$$= \frac{2y-5 \pm \sqrt{4y^2+25-20y-9y^2+30y-30}}{3}$$

$$= \frac{2y-5 \pm \sqrt{-5y^2+10y-5}}{3} = \frac{2y-5 \pm \sqrt{-5(y-1)^2}}{3}$$

$y=1$

$x = \frac{-3}{3} = -1$

$x=-1$ & $y=1$.

$\Downarrow$

No of ordered pair $= \boxed{1}$

A

Que

One of the roots of the equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form of $\frac{m+\sqrt{n}}{r}$, where m is non-zero integer and r and n are relatively prime natural number, then find $\frac{m+n+r}{4}$.

Sol

$$2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$$

$$\Rightarrow 2(10^3 \cdot x^6 - 1) + x \cdot (10^2 x^4 + 10 \cdot x^2 + 1) = 0$$

$$\Rightarrow 2((10 \cdot x^2)^3 - 1) + x(\quad '' \quad) = 0$$

$$\Rightarrow 2((10x^2 - 1)(100x^4 + 10x^2 + 1)) + x(100x^4 + 10x^2 + 1) = 0$$

$$\Rightarrow \underbrace{(100x^4 + 10x^2 + 1)}_{\times} \cdot (20x^2 + x - 2) = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1+160}}{2 \cdot 20}$$

$$= \frac{-1 \pm \sqrt{161}}{40}$$

$$= \frac{-1 + \sqrt{161}}{40}, \quad —$$

$m = -1$

$n = 161$

$+\ r = 40$

$m+n+r = 200$

$$\frac{m+n+r}{4} = \frac{200}{4} = \boxed{50}$$

Ans

Column - Matching

Column-I

A Number of triplets (x, y, z) of positive integers satisfying $2^x + 2^y + 2^z = 2336$, is

B Number of ordered pair of positive integers (x, y) satisfying $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{20}}$, is

C Number of non-negative integers 'n', such that $n+9$, $16n+9$ and $27n+9$, all are perfect squares of some integers, is

Column-II

P	1
Q	2
R	3
S	6
T	None of P, Q, R, S.

$C \longrightarrow Q.$

$B \longrightarrow R.$

$A \longrightarrow S$

(A) $2^x + 2^y + 2^z = 2336.$

Let $x \geq y \geq z.$

$x = 11 \Rightarrow 2^x = 2048$

$2048 + 2^y + 2^z = 2336.$

$2^y + 2^z = 288$

$y = 8 \rightarrow 2^z = 288 - 256$

$= 32 \Rightarrow z = 5$

$y = 7 \rightarrow$ ✗

$x = 10. \Rightarrow 2^x = 1024$

$1024 + 2^y + 2^z = 2336$

$2^y + 2^z = 1312$

$y = 10 \rightarrow 2^z = 1312 - 1024$

$= 288$ ✗

$y = 9 \rightarrow 2^y = 1024$ ✗

$$\begin{array}{r} 2336 \\ -1024 \\ \hline 1312 \end{array}$$

$$\begin{array}{r} 1312 \\ -1024 \\ \hline 288 \end{array}$$

$$\begin{array}{r} 2336 \\ 2048 \\ \hline 288 \end{array}$$

$(11, 8, 5)$

2

4

8

16

32

64

$2^7 \rightarrow 128$

$2^8 \rightarrow 256$

$2^9 \rightarrow 512$

$2^{10} \rightarrow 1024$

$2^{11} \rightarrow 2048$

4096.

$(x, y, z) = \{(11, 8, 5), (11, 5, 8), \ldots\}$

6

(B) $$\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{20}} = \frac{1}{2\sqrt{5}}$$

$$\frac{1}{\sqrt{x}} = \frac{1}{2\sqrt{5}} - \frac{1}{\sqrt{y}}$$

$()^2$

$$\frac{1}{x} = \frac{1}{20} + \frac{1}{y} - \frac{1}{\sqrt{5y}}$$

$$\begin{array}{cccc} ✓ & ✓ & ✓ & \Downarrow \\ \mathbb{Q} & \mathbb{Q} & \mathbb{Q} & \mathbb{Q} \\ & & & \Downarrow \\ & & & y = 5I_1^2 \end{array}$$

Using Symt.

$$x = 5 \cdot I_2^2$$

$$\frac{1}{\cancel{\sqrt{5}}\, I_2} + \frac{1}{\cancel{\sqrt{5}}\, I_1} = \frac{1}{2\cancel{\sqrt{5}}}$$

$$\frac{1}{I_2} + \frac{1}{I_1} = \frac{1}{2}$$

$$2I_1 + 2I_2 = I_1 I_2$$

$$I_1 I_2 - 2I_1 - 2I_2 = 0$$

$$(I_1 - 2)\cdot(I_2 - 2) = 4$$

1	$4 \Rightarrow I_1 = 3,\ I_2 = 6$ ✓
−1	$-4 \Rightarrow I_1 = 1,\ I_2 = -2$ ✗
2	$2 \Rightarrow I_1 = 4,\ I_2 = 4$ ✓
−2	$-2 \Rightarrow I_1 = 0,\ I_2 = 0$ ✗

(3, 6)
(6, 3)
(4, 4)

$\boxed{C}$

$n+9, \quad 16n+9, \quad 27n+9$

$\llcorner I_1^2 \quad \llcorner I_2^2 \quad \llcorner I_3^2$

$n = I_1^2 - 9$

$$16I_1^2 - I_2^2 = \cancel{16n} + 16\cdot 9 - (\cancel{16n} + 9)$$

$$(4I_1 - I_2)\cdot(4I_1 + I_2) = 9 \times 15 = 3^3 \cdot 5$$

$1 \quad 135 \Rightarrow 8I_1 = 136 \Rightarrow I_1 = 17 \Rightarrow$

$I_2 = 67$

$3 \quad \cancel{45}$

$5 \quad 27$

$\Rightarrow 8I_1 = 48$

$I_1 = 6$

$\Rightarrow 8I_1 = 32 \Rightarrow I_1 = 4 \times$

$$I_3^2 = 9(3n+1)$$

$$= 9\cdot(3(I_1^2 - 9) + 1)$$

$$= 9(3I_1^2 - 26)$$

$$= 9(3\cdot 17^2 - 26) \checkmark$$

$$I_3^2 = 9(3 \times 36 - 26)$$

$$\begin{array}{r} 108 \\ 26 \\ \hline 82 \end{array}$$

$$\begin{array}{r} 289 \\ \times 3 \\ \hline 867 \\ -26 \\ \hline 841. \end{array}$$

$$(4I_1 - I_2)(4I_1 + I_2) = 3^3 \cdot 5$$

$$9 \qquad 15 \longrightarrow 8I_1 = 24 \Rightarrow I_1 = 3.$$

$I_1 = 3, 17$

$\Rightarrow n + 9 = I_1^2$

$n = I_1^2 - 9$

$= 0, \; 17^2 - 9$